

Graded Rings (Product & Powers) ^①

Let $S = k[x_1, \dots, x_n]$ be a standard graded polynomial ring over a field k .

$$I \subsetneq S \quad (\text{Proper homogen. ideal})$$

$$m = (x_1, \dots, x_n) \quad (\text{homogeneous maximal ideal})$$

$$R = S/I \quad (\text{quotient ring})$$

Recall that the Poincaré series of k over R is

$$P_R^R(t) = \sum_{i \geq 0} \beta_i^R(k) t^i$$

Some established the following coefficient with u.B.

$$P_R^R(t) \leq \frac{(1+t)^n}{1-t(P_R^S(t)-1)}$$

$$\text{when, } P_R^S(t) = \sum_{i \geq 0} \beta_i^S(k) t^i$$

(2)

(1962) Golod characterized when Serre inequality becomes equality:

R is Golod $\iff K^R$ admits trivial Massey operations.

will define now.

2. Trivial Massey Operations.

Let $K = K^R$ denote the Koszul Complex of R , & let

$$H_{\geq 1}(K) = \bigoplus_{i \geq 1} H_i(K)$$

Choose homogeneous K -basis B of $H_{\geq 1}(K)$

Definition:-

(3)

A trivial Mayan operation is a function

$$\mu: \bigsqcup_{p \geq 1} B^p \longrightarrow K$$

s.t.:

① For any $h \in B$, $\mu(h)$ is a cycle representing h :-

$$d\mu(h) = 0, \quad [\mu(h)] = h.$$

② For any $p \geq 2$ & $h_1, \dots, h_p \in B$.

we can choose $\mu(h_1, \dots, h_p)$ s.t.

$$d\mu(h_1, \dots, h_p) = \sum_{j=1}^{p-1} \overline{\mu(h_1, \dots, h_j)} \mu(h_{j+1}, \dots, h_p)$$

$$\overline{a} = (-1)^{|a|+1} a.$$

Remark:- $p=2$

$$d \mu(n_1, h_2) = \overline{\mu(n_1)} \mu(h_2)$$

$\Rightarrow$ The product of any two positive Koszul homology class is zero.

$$H_{\geq 1}(K) \cdot H_{\geq 1}(K) = 0$$

How ever in general the converse is not true.

Example:-

$$\textcircled{1} R = k[x, y] / (x^2, xy)$$

$$K = K^R(x, y)$$

$$\text{A basis:- } H_1(K) = \langle [xe_x], [ye_x] \rangle$$

⑤

$$H_2(K) = \langle [xe_{xy}] \rangle$$

$$H_3(K) = 0$$

$$B = \left\{ \underset{\substack{\parallel \\ a_1}}{[xe_x]}, \underset{\substack{\parallel \\ a_2}}{[ye_x]}, \underset{\substack{\parallel \\ a_3}}{[xe_{xy}]} \right\}$$

$$\mu(a_1) = xe_x$$

$$\mu(a_2) = ye_x$$

$$\mu(a_3) = xe_{xy}$$

$$\mu(b) = 0$$

$$\forall b \in \bigcup_{p \geq 2} B^p$$

② Let $R = k[x, y, z] / (xy, xz, yz)$

$$K = K^R(x, y, z)$$

A basis for Koszul homology,

(6)

$$H_1(K) = \langle [ye_x], [ze_x], [ze_y] \rangle$$

$$H_2(K) = \langle [ze_{xy}], [xe_{yz}] \rangle$$

$$H_3(K) = 0$$

$$B = \left\{ [ye_x], \overset{a_1}{[ze_x]}, \overset{a_2}{[ze_y]}, [ze_{xy}], [xe_{yz}] \right\}$$

$$\begin{aligned} d\mu(a_1, a_2) &= \overline{\mu(a_1)} \mu(a_2) \\ &= z^2 e_{xy} \end{aligned}$$

$$\Rightarrow \mu(a_1, a_2) = z e_{xyz}$$

$$\nexists \mu(b) = 0 \quad \forall b \in \bigcup_{p \geq 2} B^p - \{a_1, a_2\}$$

$$\Rightarrow \cancel{k(x, y)}_{(x^2, xy)} \text{ in } \text{Colod.}$$

⑦

Some examples of Non Gorenstein rings are.

- Complete intersections of codimension at least 2.

or

- Non-hypersurface Gorenstein rings.

In these cases the positive Koszul homology has non-trivial multiplicative structure.

Question:- Which naturally occurring ideals give Gorenstein quotients?

Two particular important conditions are

$$I^n \quad \& \quad IJ$$

Let's look at the literature.

9

Some Open Question.

Conjecture 1:- Let $S = k[x_1, \dots, x_n]$, $\text{char}(k) = p > 0$

For a proper homogeneous ideal $I \subset S$, is

S/I^k G.O.L.D $\forall k \geq 2$?

Conjecture 2:- Let $S = k[x_1, x_2, x_3]$.

If $I, J \subsetneq S$ are arbitrary proper homogeneous ideals, is S/IJ G.O.L.D?

A different picture....

I^k & IJ

• What about $I^k J$?

• Show that $\forall I, J \exists k \gg 0$ s.t.
 $I^k J$ is G.O.L.D.

Natural extension of a result of Abhyankar

Lemma 2.1 Let I & L be ideals of S
s.t. $L^2 \subseteq I \subseteq L$. Suppose that the

map $\text{Tor}_i^S(S/I, K) \rightarrow \text{Tor}_i^S(S/L, K)$,

induced by the natural surjection $S/I \rightarrow S/L$ is
zero $\forall i \geq 1$. Then the ideal I is Golod.

$$K \div K = K^R(x_1, \dots, x_n), R = S/I.$$

$$\beta \text{ a basis } H_{\geq 1}(K),$$

μ can be chosen s.t.

$$\mu(h_{\lambda_1})\mu(h_{\lambda_2}) = 0 \forall h_{\lambda_1}, h_{\lambda_2} \in \beta$$

$$\& \mu(h_{\lambda_1}, \dots, h_{\lambda_n}) = 0 \quad \forall n \geq 2.$$

(11)

We note that the map

$$\mathrm{Tor}_i^S(S/I, K) \longrightarrow \mathrm{Tor}_i^S(S/L, K)$$

can be identified with the map.

$$\textcircled{2} \quad H_i(K) \longrightarrow H_i(K/LK)$$

induced by natural projection, $K \longrightarrow K/LK$.

For each $h_x \in B$, the vsm of $\textcircled{2}$

$$\Rightarrow h_x = [z_x] \text{ for some } z_x \in Z(K) \cap LK$$

Now we define, $\mu(h_x) = z_x$.

The result $L^2 \subseteq I \Rightarrow \mu(h_{x_1})\mu(h_{x_2}) = 0$

$\forall h_{x_1}, h_{x_2} \in B$.

(12)

Proposition: Let I & J be ideal in S . There exists an integer d_0 s.t. $\forall d \geq d_0$, the natural map

$$\mathrm{Tor}_i^S(S/J^d I, K) \longrightarrow \mathrm{Tor}_i^S(S/J^{d-1} I, K)$$

induced by the surjection $S/J^d I \rightarrow S/J^{d-1} I$ is zero $\forall i > 0$.
 $\left(\begin{array}{l} Z = \ker(K) \\ B = \mathrm{im}(K) \end{array} \right), K = K \otimes_S^L S$

pf: Consider, $H_*(J^d I \otimes_S K) \longrightarrow H_*(J^{d-1} I \otimes_S K)$

we have,

$$\ker(J^d I \otimes_S K_i \rightarrow J^d I \otimes_S K_{i-1}) = Z_i \cap J^d I K_i$$

$$\mathrm{im}(J^{d-1} I \otimes_S K_i \rightarrow J^d I \otimes_S K_{i-1}) = J^{d-1} I Z_i$$

Ardin-Rees $\Rightarrow \exists d_0$ s.t. $\forall d \geq d_0$,

$$J^d I K_i \cap Z_i = J^{d-d_0} (J^{d_0} I K_i \cap Z_i)$$

$\forall i \geq 0$.

(13)

Since $H_i(J^{d_0} IK)$ is a k -vector space

$$J(J^{d_0} IK \cap Z_i) \subseteq B_i(J^{d_0} IK) \\ \parallel \\ J^{d_0} IK \cap Z_i$$

$\Rightarrow$

$$J^d IK_i \cap Z_i =_{(a.e)} J^{d-d_0-1} J(J^{d_0} IK_i \cap Z_i)$$

$$\subseteq J^{d-d_0-1} (J^{d_0} IK_i) = J^{d-1} IK_i$$

End.